

SCIENCE IS THE PATH TO KNOWLEDGE, AND FINDING OUT ABOUT HOW OUR UNIVERSE WORKS. YOU CAN'T BE A GEEK AND NOT KNOW YOUR SCIENCE!

THIS MONTH IN SCIENCE:

This month, learn about mysterious sailing stones in Earth, discover desalination technologies in Cutting Edge, learn how neurotransmitters function in Health, and more.



ResMem

Researchers have created a model known as ResMem that gives a memorability score for images. It is a software that tells you how likely it is that people can recall the photos you are uploading to social media. <https://digit.in/resmem>

WHAT'S NEW

Blinking Giant discovered

Astronomers have come across a first of its kind star with a mysterious "orbital companion" capable of eclipsing the star for a period of months, every few decades, as seen from the Earth. The entire system is called the "blinking giant", and researchers believe that the orbital companion is likely a planet or another star, one that is surrounded by a disc. VVV-WIT-07 is one of the variable stars, such as Tabby's star (aka the WTF star), but shows a flux of 80 percent as compared to the 22 percent seen in Tabby's star. VVV-WIT-07 is pretty unusual, which was one of the hints that the researchers had towards guessing the nature of the "occultor". By estimating the possible shape, opacity and velocity



of the object that made the star blink, researchers came to the conclusion that this was a unique system. The "WIT" in the designation

stands for "what-is-this?", the scientists have yet to discover exactly what the companion of the object is. The researchers also checked if some random dark body between the Earth and the star was the cause of the blinking, but for this to have happened, there would have to be an impossibly large number of dark bodies randomly floating about in the Milky Way. <https://digit.in/discworld>

Beetle inspired condenser

The darkling beetles, found in the Namib Desert, shed excess heat in the form of infrared radiation. This reduces its body temperature below the dew point, causing water vapour to condense on its back. It positions itself on a ridge of sand, extends its legs, angles its body, and appears to catch the fog. A series of grooves and ridges then direct this water towards its mouth. Scientists use the same principle for dew



condensers, which only work at night as the Sun gives out more heat than the condensers can shed. Till now. Researchers from the University of Buffalo have used cheap, widely available materials to make a

dew condenser using a thin film of a material known as polydimethylsiloxane (PDMS) that is very efficient at thermal radiation, and backing it with silver which reflects the sunlight. The new radiative vapour condensation technology allows for water harvesting during the day time, without the use of a power source. Previously the team had worked on developing solar evaporators. <https://digit.in/dlb>



Deep brain stimulation

Researchers have developed algorithms that could one day be used for direct stimulation of brain cells to reduce tremors associated with Parkinson's disease, or even to help relieve chronic pain. The adaptive DBS system simultaneously senses brain signals while stimulating it. <https://digit.in/dbs>



Tracking dust

Researchers have tracked a grain of asteroid dust on its journey since the creation of the solar system, by combining sample analysis with simulations. This particular grain of dust is 4.5 billion years old, and is thought to have been among the earliest solids to have formed in the system. <https://digit.in/tdust>



Nanodecoys tackle COVID

Instead of directly tackling the virus itself, researchers have used a new approach that uses nanodecoys, analogs of human lung cells, to make the virus attack the decoys, instead of the cells. Mimicking the target is an approach that can potentially be viable on novel viruses. <https://digit.in/nanodecoys>